Welsh ⇄ English Machine Translation Benchmark

FLORES-200 devtest · 100 sentences per direction · 22 October 2026

Prepared for BlueMesh / Netstratum

Key findings

- Best English → Welsh: **Techiaith NLLB 1.3B fine-tunes** — COMET 87.3 (good, but needs review for official text).
- Best Welsh → English: **Qwen 3.6 27B** — COMET 89.9 (near-professional).
- Deployed system **Qwen 3.6 27B** ranks 3 of 8 on mean COMET (87.7: en→cy 85.5, cy→en 89.9).
- Fastest usable system (COMET ≥ 80 both ways): **NLLB-200 3.3B** at 1.22 s per sentence (mean of both directions); the deployed system averages 1.60 s.
- Measured GPU memory: Techiaith NLLB 1.3B fine-tunes 3 GB; NLLB-200 3.3B 7 GB; Qwen 3.6 27B 22 GB; TranslateGemma 27B 32 GB; Gemma 4 31B-it 32 GB; Gemma 4 31B-it (Q4) 25 GB; TranslateGemma 12B 10 GB; TranslateGemma 4B 4.4 GB.

Recommendation: Techiaith NLLB 1.3B fine-tunes has the highest mean COMET (87.9) across both directions, 0.2 points above the deployed Qwen 3.6 27B, but its licence (CC-BY-NC-4.0) rules out commercial use. Among commercially usable systems the deployed Qwen 3.6 27B scores highest.

Systems evaluated: 8. Metrics: COMET (wmt22-comet-da), chrF++ and BLEU. All scores in this report are computed from the benchmark output files; COMET is shown on a 0–100 scale.

1. Executive summary

We compared 8 machine-translation systems on 100 sentences of the public FLORES-200 benchmark, translating in both directions between English and Welsh. Each output was compared to a professional human translation using three automatic metrics; the main one, COMET, estimates how well the meaning is preserved and correlates best with human judgement.

English → Welsh is the harder direction. Averaged over all systems, COMET is 77.9 for English → Welsh versus 87.6 for Welsh → English. Producing correct Welsh — mutations, gender, word order and idiomatic vocabulary — is where systems differ most, so it should weigh most when choosing a model for Welsh-language output.

The best English → Welsh system is **Techiaith NLLB 1.3B fine-tunes** (87.3); the best Welsh → English system is **Qwen 3.6 27B** (89.9). Across both directions **Techiaith NLLB 1.3B fine-tunes** has the highest mean COMET (87.9), while the deployed system is Qwen 3.6 27B (87.7).

What the COMET numbers mean in practice

COMET has no absolute pass mark. The bands below are a *rule of thumb* for this kind of sentence-level news/encyclopaedic text, based on typical correlations with human ratings — they are not guarantees, and individual sentences can still contain serious errors at any score.

COMET	Rule-of-thumb reading	Systems in this band (direction)
≥ 88	Near-professional: fluent and accurate; occasional slips that a reviewer would catch quickly.	Techiaith NLLB 1.3B fine-tunes (cy→en 88.4); NLLB-200 3.3B (cy→en 88.8); Qwen 3.6 27B (cy→en 89.9); TranslateGemma 27B (cy→en 89.0); Gemma 4 31B-it (cy→en 88.4); Gemma 4 31B-it (Q4) (cy→en 88.6)
83 – 88	Good: meaning usually right, but wording, mutations or terminology need review before official publication.	Techiaith NLLB 1.3B fine-tunes (en→cy 87.3); NLLB-200 3.3B (en→cy 86.8); Qwen 3.6 27B (en→cy 85.5); TranslateGemma 27B (en→cy 84.3); TranslateGemma 12B (cy→en 87.1)
80 – 83	Usable for gist; frequent minor errors.	TranslateGemma 4B (cy→en 80.5)
< 80	Noticeable errors: mistranslations, untranslated or invented words; not suitable without full post-editing.	Gemma 4 31B-it (en→cy 77.2); Gemma 4 31B-it (Q4) (en→cy 76.7); TranslateGemma 12B (en→cy 76.6); TranslateGemma 4B (en→cy 48.9)

In everyday terms: a system around COMET 89 produces Welsh or English that a fluent reader would mostly accept as written; around 84 the text is understandable and largely correct, but a Welsh speaker would want to edit a noticeable share of sentences; in the high 70s readers will meet clear errors in many sentences.

Scores are from a 100-sentence sample, so differences of about 1–2 COMET points between systems should not be over-interpreted. Speed figures were measured on a shared production GPU server and are indicative only.

2. Methodology

Test set

We used the **FLORES-200 devtest** split from Meta's public release, aligned English (`eng_Latn`) and Welsh (`cym_Latn`) files of 1,012 sentences each. An evenly spaced sample of 100 sentences was taken (every 10th line, starting at line 1), and the same sentences were translated in both directions: English → Welsh (en→cy) and Welsh → English (cy→en). The FLORES texts are professionally translated sentences drawn from Wikipedia-style articles (news, science, travel, health).

Decoding and prompting

- LLMs used greedy decoding (temperature 0), one sentence per request, with “thinking”/reasoning modes disabled.
- TranslateGemma models used Google's official translation prompt template (reproduced in the appendix).
- Other LLMs used a generic system prompt asking for a professional, fluent translation that preserves meaning, tone, names and numbers, outputting only the translation.
- NLLB-based models used beam search with 4 beams and the FLORES language codes as source/target tokens.

Metrics

- **COMET** (Unbabel/wmt22-comet-da), reference-based: a neural metric built on XLM-RoBERTa (whose training languages include Welsh) that scores each translation against the source and the reference. Reported ×100; the system score is the mean of sentence scores. Primary metric for this report.
- **chrF++** (sacrebleu, corpus-level, word_order=2): character n-gram F-score plus word bigrams; more tolerant than BLEU of Welsh morphology.
- **BLEU** (sacrebleu, corpus-level, default tokenisation): word n-gram overlap with the reference.
- **Latency**: wall-clock seconds per sentence, measured as total time for the sample divided by the number of sentences, sending sequential single requests.

Hardware

All systems ran on one server with 2× NVIDIA RTX PRO 6000 Blackwell GPUs (96 GB each). The GPUs were shared with other production services during the runs, so timings are indicative rather than controlled benchmarks.

Limitations

- Sample size: 100 sentences per direction gives an uncertainty of roughly ±1–2 COMET points; small differences are not significant.
- Single reference: valid alternative translations are penalised, especially by BLEU and chrF++.
- Domain: FLORES is Wikipedia-style prose, which differs from public-service, legal or conversational text.
- BLEU penalises Welsh initial mutations and flexible word order even when the translation is correct, so it understates Welsh-output quality.
- Latency was affected by other workloads on the GPUs.
- Systems ran on different runtimes and quantisations (see Models evaluated), so quality and speed reflect the configuration tested, not the model in the abstract.

3. Models evaluated

Systems are listed in order of mean COMET across both directions. VRAM is the GPU memory observed while serving, where it was measured.

System	Type	Size	Runtime / quantisation	VRAM	Licence
Techiaith NLLB 1.3B fine-tunes techiaith/nllb-200-1.3B-ft-eng-to-cym-legislation (en→cy, legislation-domain) + techiaith/nllb-200-1.3B-ft-cym-to-eng (cy→en)	Dedicated MT (Welsh-specialised)	1.3B	HF transformers fp16	3 GB	CC-BY-NC-4.0
NLLB-200 3.3B facebook/nllb-200-3.3B	Dedicated MT (multilingual)	3.3B	HF transformers fp16	7 GB	CC-BY-NC-4.0
Qwen 3.6 27B (deployed) Qwen/Qwen3.6-27B	LLM (general)	27B	Ollama (llama.cpp) Q4_K_M	22 GB	Apache-2.0
TranslateGemma 27B google/translategemma-27b-it	LLM (translation-tuned)	27B	Ollama (llama.cpp) Q8_0	32 GB	Gemma Terms of Use
Gemma 4 31B-it google/gemma-4-31B-it	LLM (general)	31B	HF transformers bitsandbytes int8	32 GB	Apache-2.0
Gemma 4 31B-it (Q4) google/gemma-4-31B-it	LLM (general)	31B	Ollama (llama.cpp) Q4_K_M	25 GB	Apache-2.0
TranslateGemma 12B google/translategemma-12b-it	LLM (translation-tuned)	12B	Ollama (llama.cpp) Q4_K_M	10 GB	Gemma Terms of Use
TranslateGemma 4B google/translategemma-4b-it	LLM (translation-tuned)	4B	Ollama (llama.cpp) Q4_K_M	4.4 GB	Gemma Terms of Use

Licence notes

- **Gemma Terms of Use** permit commercial use subject to Google's prohibited-use policy; **Apache-2.0** is a permissive open-source licence.
- **CC-BY-NC** licences (Techiaith NLLB 1.3B fine-tunes and NLLB-200 3.3B) are **non-commercial**: these models can be used for research and evaluation but not in a commercial product or paid service without a separate licence.

4. Results

Table 1 gives corpus-level scores. Higher is better for COMET, chrF++ and BLEU; lower is better for seconds per sentence (mean of the two directions). The best value in each column is shown in bold.

System	English → Welsh			Welsh → English			Speed
	COMET	chrF++	BLEU	COMET	chrF++	BLEU	s / sent.
Techiaith NLLB 1.3B fine-tunes	**87.3**	66.4	46.3	88.4	75.4	58.7	2.32
NLLB-200 3.3B	86.8	**68.3**	**48.1**	88.8	**76.8**	**60.8**	1.22
Qwen 3.6 27B ●	85.5	60.6	35.7	**89.9**	74.6	55.0	1.60
TranslateGemma 27B	84.3	53.5	25.3	89.0	71.1	49.6	1.59
Gemma 4 31B-it	77.2	50.0	20.9	88.4	74.0	55.2	14.5
Gemma 4 31B-it (Q4)	76.7	53.4	28.1	88.6	72.6	54.0	1.65
TranslateGemma 12B	76.6	46.0	18.2	87.1	64.9	40.8	1.17
TranslateGemma 4B	48.9	26.3	5.3	80.5	52.2	28.6	**0.74**

Table 1. Corpus-level scores, ordered by mean COMET. ● marks the deployed system.

COMET by system and direction

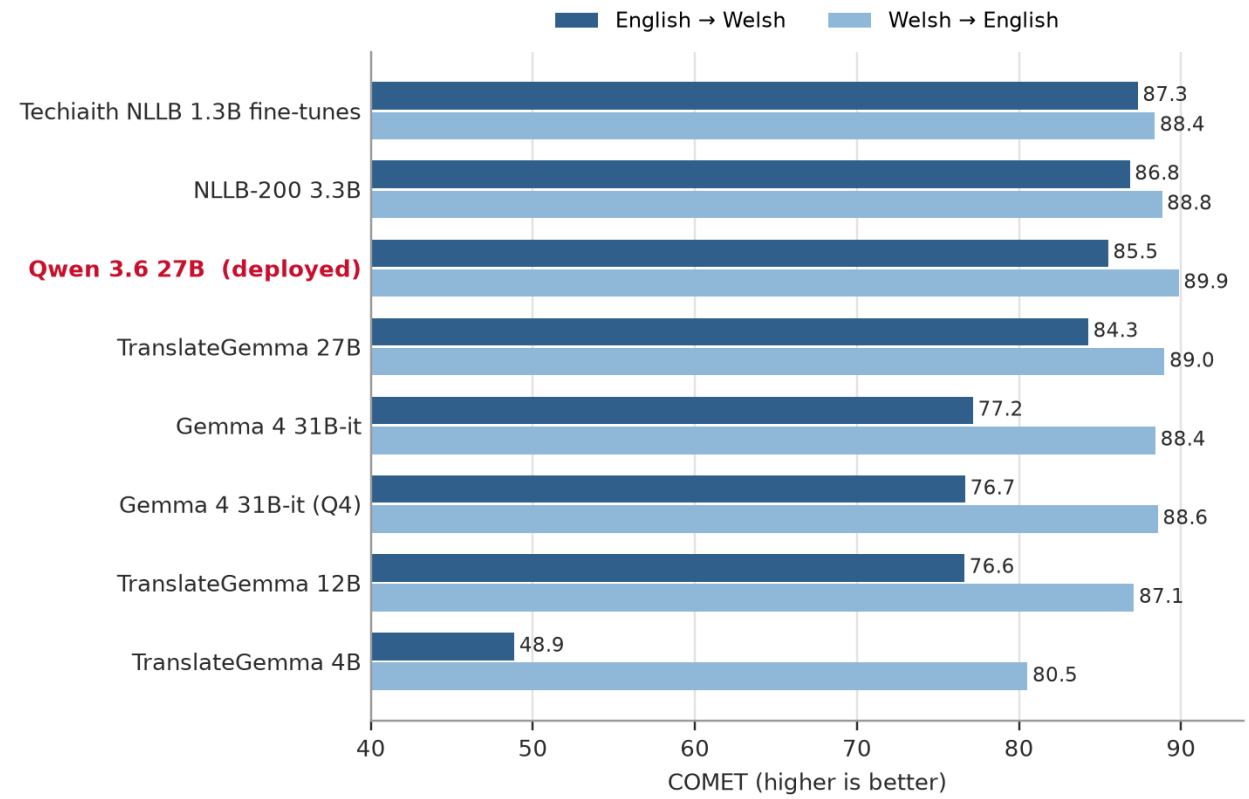
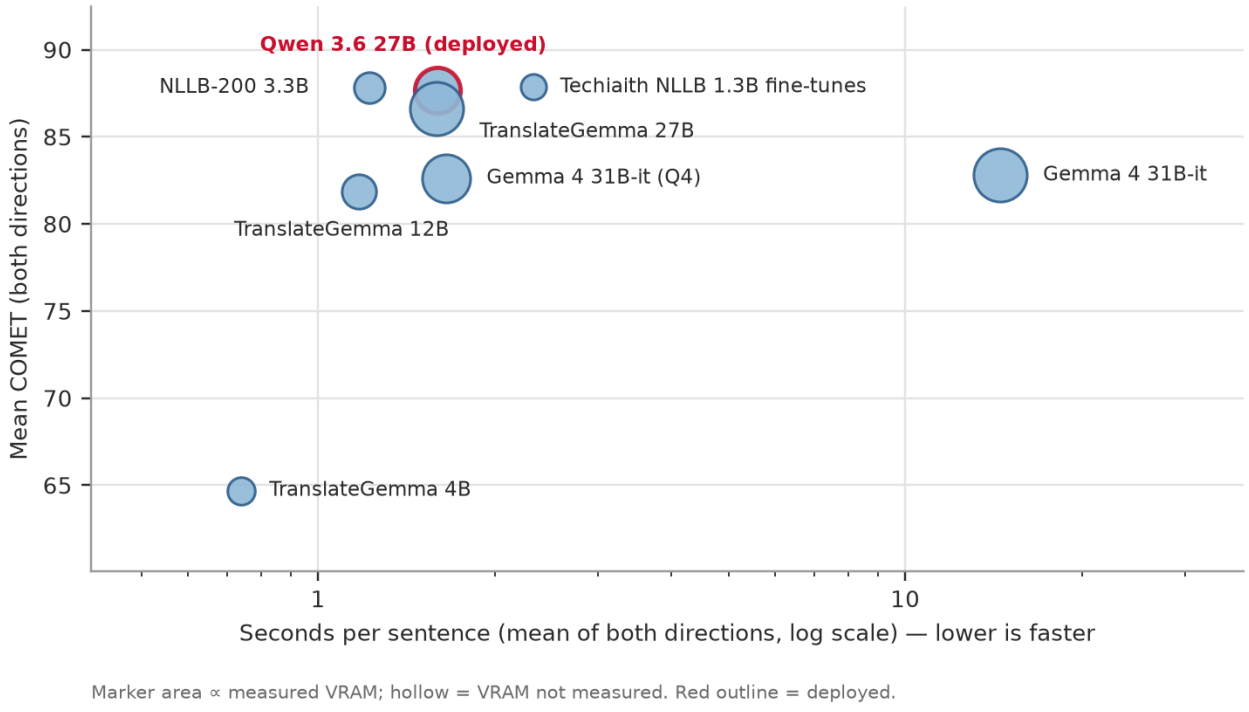


Figure 1. Corpus COMET for each system. The axis starts at 40 to make differences visible; deployed system labelled in red.

Quality vs speed



Marker area ∝ measured VRAM; hollow = VRAM not measured. Red outline = deployed.

Figure 2. Mean COMET against mean seconds per sentence. Up and to the left is better. Timings come from a shared GPU server and are indicative.

Sentence-level score distribution

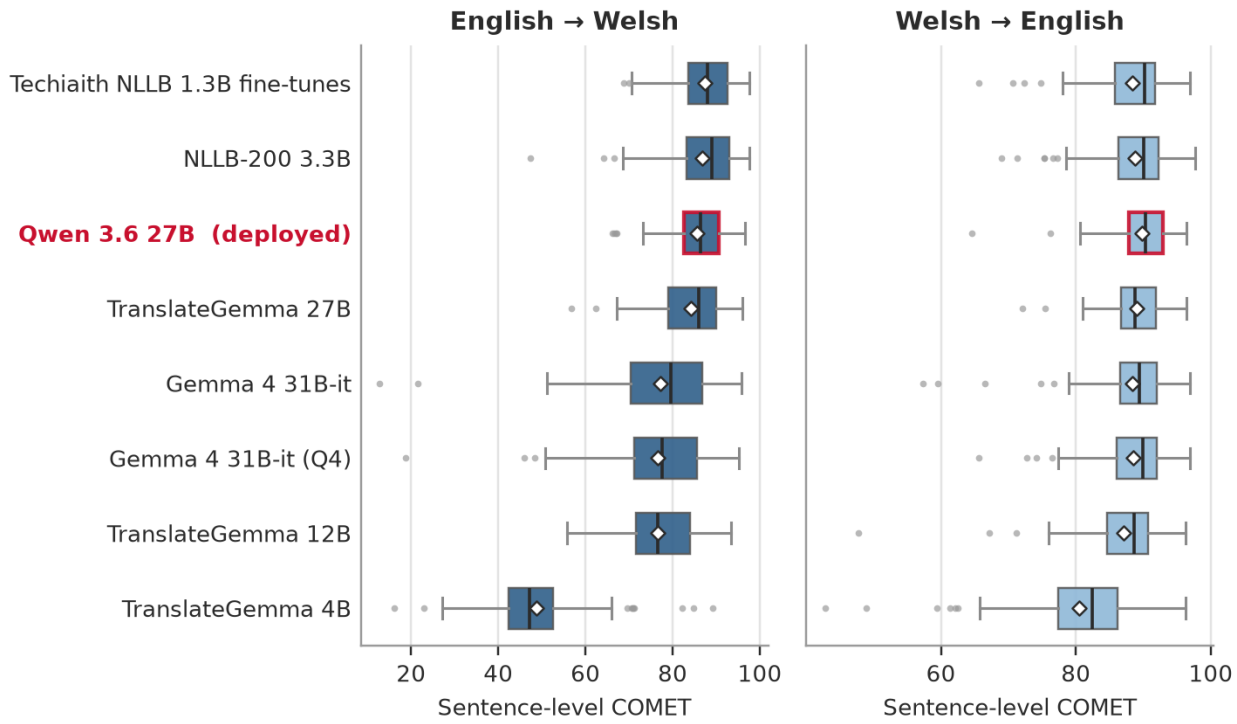


Figure 3. Distribution of per-sentence COMET. Boxes show the inter-quartile range, the line the median, the white diamond the mean, and dots individual outlying sentences. A long left tail means occasional badly wrong translations even when the average is good.

Tail behaviour

System	Direction	Median	10th pct.	Sentences < 70	Sentences ≥ 88	Runaway outputs
Techiaith NLLB 1.3B fine-tunes	en→cy	88.1	79.2	2	51	0
Techiaith NLLB 1.3B fine-tunes	cy→en	90.2	81.2	1	63	0
NLLB-200 3.3B	en→cy	88.9	77.7	5	55	0
NLLB-200 3.3B	cy→en	90.1	82.2	1	67	0
Qwen 3.6 27B	en→cy	86.4	77.0	4	39	0
Qwen 3.6 27B	cy→en	90.3	84.6	1	74	0
TranslateGemma 27B	en→cy	86.0	75.4	4	36	0
TranslateGemma 27B	cy→en	88.8	84.3	0	62	0
Gemma 4 31B-it	en→cy	79.5	61.4	24	21	2
Gemma 4 31B-it	cy→en	89.4	83.6	3	63	0
Gemma 4 31B-it (Q4)	en→cy	77.5	60.2	22	16	1
Gemma 4 31B-it (Q4)	cy→en	89.9	83.0	1	63	0
TranslateGemma 12B	en→cy	76.6	62.9	22	14	0
TranslateGemma 12B	cy→en	88.6	80.2	2	55	0
TranslateGemma 4B	en→cy	47.2	38.7	93	1	4
TranslateGemma 4B	cy→en	82.4	67.7	14	21	0

Table 2. Per-sentence COMET statistics (out of 100 sentences per direction). Runaway outputs are hypotheses more than 2.5× the reference length, usually repetition loops.

chrF++ by system and direction

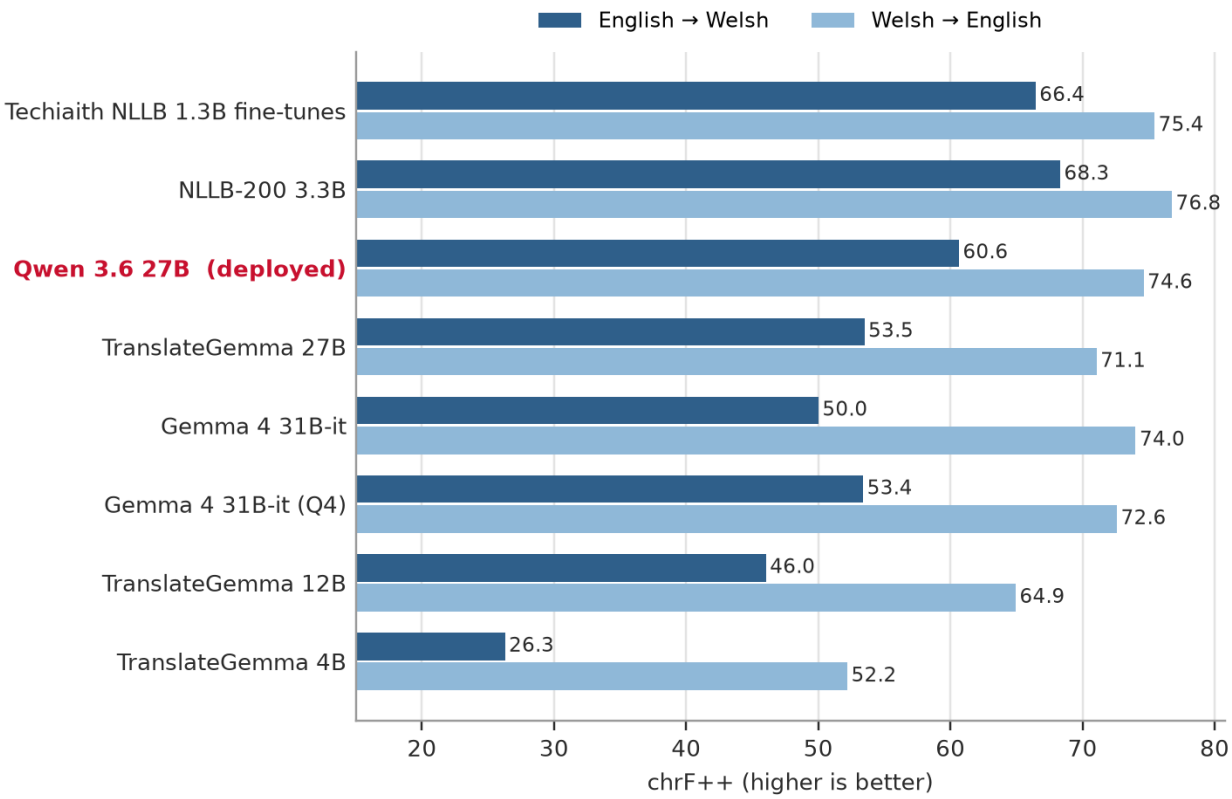


Figure 4. Corpus chrF++ (axis starts at 15). chrF++ measures character overlap with the reference and is less sensitive to meaning than COMET.

Is the difference real? (significance)

Each system is compared with the deployed Qwen 3.6 27B on the same sentences: the per-sentence COMET differences are resampled 2,000 times (paired bootstrap) to give a 95% interval. If the interval excludes zero the difference is unlikely to be sampling noise; otherwise it is reported as a tie.

System	en→cy Δ COMET [95% CI]		cy→en Δ COMET [95% CI]	
Techiaith NLLB 1.3B fine-tunes	+1.8 [+0.9, +2.8]	**better**	-1.5 [-2.3, -0.7]	**worse**
NLLB-200 3.3B	+1.3 [+0.0, +2.6]	**better**	-1.0 [-1.7, -0.4]	**worse**
TranslateGemma 27B	-1.2 [-2.4, -0.0]	**worse**	-0.9 [-1.3, -0.4]	**worse**
Gemma 4 31B-it	-8.3 [-11.1, -5.9]	**worse**	-1.4 [-2.4, -0.6]	**worse**
Gemma 4 31B-it (Q4)	-8.8 [-11.2, -6.8]	**worse**	-1.3 [-2.0, -0.6]	**worse**
TranslateGemma 12B	-8.9 [-10.7, -7.1]	**worse**	-2.8 [-3.7, -1.9]	**worse**
TranslateGemma 4B	-36.6 [-39.1, -34.3]	**worse**	-9.3 [-11.1, -7.8]	**worse**

Table 3. COMET difference versus Qwen 3.6 27B (positive = scored higher than the deployed system).

5. Analysis by direction

5.1 English → Welsh

Ranking by COMET: 1. Techiaith NLLB 1.3B fine-tunes (87.3); 2. NLLB-200 3.3B (86.8); 3. Qwen 3.6 27B (85.5); 4. TranslateGemma 27B (84.3); 5. Gemma 4 31B-it (77.2); 6. Gemma 4 31B-it (Q4) (76.7); 7. TranslateGemma 12B (76.6); 8. TranslateGemma 4B (48.9).

- **Techiaith NLLB 1.3B fine-tunes** has the highest COMET (87.3); **TranslateGemma 4B** the lowest (48.9). The spread between best and worst is 38.5 COMET points.
- The top two systems (Techiaith NLLB 1.3B fine-tunes and NLLB-200 3.3B) are only 0.5 COMET apart, which is within the sampling uncertainty of a 100-sentence test set; treat them as statistically tied on this evidence.
- The metrics do not agree on the winner: the best chrF++ (NLLB-200 3.3B, 68.3) and BLEU (NLLB-200 3.3B, 48.1) differ from the COMET leader. String-overlap metrics reward wording that matches the single reference, whereas COMET estimates meaning preservation, so this usually indicates that outputs differ in phrasing rather than in accuracy.
- Translation-tuned vs general LLMs: the best translation-tuned model (TranslateGemma 27B, 84.3) vs the best general LLM (Qwen 3.6 27B, 85.5) — the general-purpose model leads by 1.2 COMET.
- Dedicated MT vs LLMs: best dedicated system Techiaith NLLB 1.3B fine-tunes (87.3) vs best LLM Qwen 3.6 27B (85.5); the dedicated MT model is ahead by 1.8 COMET.
- Model size within TranslateGemma: TranslateGemma 4B 48.9; TranslateGemma 12B 76.6; TranslateGemma 27B 84.3. Going from 4B to 27B gains 35.4 COMET in this direction (note the smaller models ran at a different quantisation).
- Quantisation of google/gemma-4-31B-it: bitsandbytes int8 (HF transformers): 77.2; Q4_K_M (Ollama (llama.cpp)): 76.7. The difference is 0.5 COMET; because runtime and quantisation changed together, this cannot be attributed to quantisation alone.
- Runaway outputs — text more than 2.5× the reference length, typically a word or phrase repeated until the generation limit: Gemma 4 31B-it 2 (COMET 12.7, 21.5); Gemma 4 31B-it (Q4) 1 (COMET 18.8); TranslateGemma 4B 4 (COMET 27.5, 22.9, 27.2, 16.0). These failures are rare but severe, and a production deployment would need a length guard.
- Speed: TranslateGemma 4B was fastest at 0.92 s per sentence and Gemma 4 31B-it slowest at 22.3 s (single sequential requests on a shared GPU server; indicative only).

5.2 Welsh → English

Ranking by COMET: 1. Qwen 3.6 27B (89.9); 2. TranslateGemma 27B (89.0); 3. NLLB-200 3.3B (88.8); 4. Gemma 4 31B-it (Q4) (88.6); 5. Gemma 4 31B-it (88.4); 6. Techiaith NLLB 1.3B fine-tunes (88.4); 7. TranslateGemma 12B (87.1); 8. TranslateGemma 4B (80.5).

- **Qwen 3.6 27B** has the highest COMET (89.9); **TranslateGemma 4B** the lowest (80.5). The spread between best and worst is 9.3 COMET points.
- The top two systems (Qwen 3.6 27B and TranslateGemma 27B) are only 0.9 COMET apart, which is within the sampling uncertainty of a 100-sentence test set; treat them as statistically tied on this evidence.
- The metrics do not agree on the winner: the best chrF++ (NLLB-200 3.3B, 76.8) and BLEU (NLLB-200 3.3B, 60.8) differ from the COMET leader. String-overlap metrics reward wording that matches the single reference, whereas COMET estimates meaning preservation, so this usually indicates that outputs differ in phrasing rather than in accuracy.

- Translation-tuned vs general LLMs: the best translation-tuned model (TranslateGemma 27B, 89.0) vs the best general LLM (Qwen 3.6 27B, 89.9) — the general-purpose model leads by 0.9 COMET.
- Dedicated MT vs LLMs: best dedicated system NLLB-200 3.3B (88.8) vs best LLM Qwen 3.6 27B (89.9); the LLM is ahead by 1.0 COMET.
- Model size within TranslateGemma: TranslateGemma 4B 80.5; TranslateGemma 12B 87.1; TranslateGemma 27B 89.0. Going from 4B to 27B gains 8.5 COMET in this direction (note the smaller models ran at a different quantisation).
- Quantisation of google/gemma-4-31B-it: Q4_K_M (Ollama (llama.cpp)): 88.6; bitsandbytes int8 (HF transformers): 88.4. The difference is 0.1 COMET; because runtime and quantisation changed together, this cannot be attributed to quantisation alone.
- Speed: NLLB-200 3.3B was fastest at 0.50 s per sentence and Gemma 4 31B-it slowest at 6.72 s (single sequential requests on a shared GPU server; indicative only).

These observations are generated directly from the scores. With a 100-sentence sample, differences under about 1–2 COMET points should be read as ties.

5.3 Caveats from manual spot checks

- **Possible home advantage for NLLB-based systems.** FLORES-200 was built by Meta's NLLB team and its dev split was used when developing NLLB, so NLLB-200 and the Techiaith fine-tunes built on it may match the test set's style more closely than the LLMs do. Their lead should be confirmed on in-domain text.
- **Techiaith English → Welsh model is legislation-domain.** The general-domain English → Welsh fine-tune on Hugging Face is missing part of its weights, so the legislation fine-tune (techiaith/nllb-200-1.3B-ft-eng-to-cym-legislation) was used instead.
- **High scores still hide real errors.** In spot checks, NLLB-200 rendered “dinosaur feathers” as “rhewiau'r ddinasoedd” (“the cities' frosts”) while scoring well overall, and Qwen 3.6 left English technical terms untranslated (“barbs a barbules”).
- **TranslateGemma 4B is not usable for English → Welsh.** Its output contains invented words and occasionally another language (e.g. Dutch “Omdat”).
- **Gemma 4 31B (int8) produced runaway outputs** (long repetition loops) on two English → Welsh sentences, which also explains its high average latency.

6. Qualitative examples

For each direction we show sentences where the systems disagree most (largest spread of sentence-level COMET) and one sentence that every system handles well. Each output is followed by its sentence COMET. Line numbers refer to the FLORES-200 devtest file (1-based).

6.1 English → Welsh

Example 1.1 — devtest line 451 (large disagreement between systems)

Source	Ancient China had a unique way of showing different time periods; each stage of China or each family that was in power was a distinctive dynasty.
Reference	Roedd gan Tsieina hynafol ffordd unigryw o ddangos cyfnodau amser gwahanol; roedd pob cyfnod o Tsieina neu bob teulu oedd mewn grym yn frenhinlin neilltuol.
TranslateGemma 12B COMET 89.4	Roedd gan Tsieina hynaf dull arbennig o ddangos cyfnodau amser gwahanol; pob cyfnod o hanes Tsieina neu bob teulu a oedd mewn grym oedd yn deyrnasiad arbennig.
Qwen 3.6 27B COMET 88.3	Roedd Tsieina hynafol yn defnyddio dull unigryw o ddangos cyfnodau amser gwahanol; roedd pob cam o Tsieina neu bob teulu oedd mewn pŵer yn frenhiniaeth wahanol.
NLLB-200 3.3B COMET 87.7	Roedd gan hen Tsieina ffordd unigryw o ddangos gwahanol gyfnodau amser; roedd pob cam o Tsieina neu bob teulu a oedd mewn grym yn deyrnas nodweddiadol.
Techiaith NLLB 1.3B fine-tunes COMET 86.3	Roedd gan Hen Tsieina ffordd unigryw o ddangos cyfnodau amser gwahanol; roedd pob cyfnod o Tsieina neu bob teulu a oedd yn y pŵer yn deddfiad nodweddiadol.
TranslateGemma 27B COMET 83.3	Roedd gan Tsieina gyfredinol fodd unigryw o ddangos cyfnodau amser gwahanol; bob cam o hanes Tsieina neu bob teulu a oedd mewn grym oedd yn dynasti unigol.
TranslateGemma 4B COMET 51.8	[[China fel yr oedd, oedd yn boddi ar ffordd unigol o ddeunyddio cyfnodau gwir, ac ychwanodd bob cyfnod neu bob teulu oedd yn llywodraeth, dynasti unigol.
Gemma 4 31B-it (Q4) COMET 50.8	Roedd gan Llwysgeithiaeth Gynulleidfaethol ffordd unigryw o ddangos cyd-cyfnodau gwahanol; roedd pob cam o Llwysgeithiaeth neu bob teulu oedd yn y pwer yn dynasti dynbygol.
Gemma 4 31B-it COMET 12.7	Roedd gan Gynhanedd Gobaith ffordd unigryw o ddangos cydfresau amser gwahanol; roedd pob cam o hanes llwybr llwybr llwybr llwybr llwybr llwybr llwybr llwybr llwybr llwybr llwybr llwybr llwybr llwybr llwybr llwybr llwybr llwybr llwybr [… output truncated for display; 4,793 characters in total]

Example 1.2 — devtest line 481 (large disagreement between systems)

Source	The High Middle Ages were preceded by the Early Middle Ages and followed by the Late Middle Ages, which by convention ends around 1500.
Reference	Cafodd yr Oesoedd Canol Uchel eu rhagflaenu gan yr Oesoedd Canol Cynnar a'u dilyn gan yr Oesoedd Canol Hwyr, sydd yn ôl traddodiad yn gorffen tua 1500.
TranslateGemma 27B COMET 86.2	Rhagflaenwyd yr Oesoedd Canol Uwch gan yr Oesoedd Canol Cynharach, ac yna ddaeth yr Oesoedd Canol Diweddar, sydd, trwy drefn, yn dod i ben tua 1500.
Qwen 3.6 27B COMET 84.6	Cafodd yr Oesoedd Canol Canol eu blaenoriaethu gan yr Oesoedd Canol Cynnar a'u dilyn gan yr Oesoedd Canol Hwyr, sy'n gorffen tua 1500 yn ôl arfer.
Techiaith NLLB 1.3B fine-tunes COMET 82.9	Cynnarwyd yr Uchel Oesoedd Canol gan yr Oesoedd Canol Cynnar ac yna'r Oesoedd Canol Hwyr, sy'n dod i ben o gwmpas 1500 yn ôl y confensiwn.
Gemma 4 31B-it COMET 79.0	Fe ddaeth yr Oesau Canol Cynnar cyn yr Oesau Canol Uchel, a ddaeth yr Oesau Canol Diweddar wedyn, a thyrant i ben tua 1500 yn ôl y arafiad.
NLLB-200 3.3B COMET 77.7	Cynodd yr Oesoedd Canol Uwch yr Oesoedd Canol Gynnar a dilynwyd gan yr Oesoedd Canol Hwyr, sy'n gorffen tua 1500.
TranslateGemma 12B COMET 65.9	Cafodd y Canol Oes Gyfnach safleiniad gan y Canol Oes Cynnar, ac wedi iddi daeth y Canol Oes Diweddaraf, a mae'n arferol ystyried bod hi wedi dod i ben tua'r flwyddyn 1500.
Gemma 4 31B-it (Q4) COMET 48.3	Fe ddaeth y Canolynnesau Cynnar cyn y Canolynnesau Uchel, a ddaeth y Canolynnesau Diweddar wedyn, a thybiau, yn dod i ben tua 1500.
TranslateGemma 4B COMET 16.0	Y Tyddyn Fach oedd y Tyddyn Fach, a gefinwyd gan Tyddyn Fach, a chynhirwyd gan Tyddyn Fach, oedd yn diweddáu dwylo ddiwedd, a'r diwedd oedd dwylo dwylo dwylo dwylo dwylo dwylo dwylo dwylo dwylo dwylo dwylo dwylo dwylo dwylo dwylo dwylo dwylo dwylo dwylo [… output truncated for display; 1,061 characters in total]

Example 1.3 — devtest line 51 (all systems score well)

Source	The Iraq Study Group presented its report at 12.00 GMT today.
Reference	Cyflwynodd Grŵp Astudio Irac ei adroddiad am 12.00 GMT heddiw.
Techiaith NLLB 1.3B fine-tunes COMET 97.7	Cyflwynodd Grŵp Astudio Irac ei adroddiad am 12.00 GMT heddiw.
NLLB-200 3.3B COMET 95.1	Cyflwynodd y Grŵp Astudio Irac ei adroddiad am 12.00 GMT heddiw.
Qwen 3.6 27B COMET 94.4	Cyflwynodd Grŵp Astudiaeth Irag ei adroddiad am 12.00 GMT heddiw.
TranslateGemma 27B COMET 94.3	Cyflwynodd Grŵp Astudiaeth Iraq ei adroddiad am 12.00 GMT heddiw.
Gemma 4 31B-it (Q4) COMET 94.1	Cyflwynodd Grŵp Astudiaeth yr Iraq ei adroddiad am 12.00 GMT heddiw.
TranslateGemma 12B COMET 92.7	Cyflwynodd Grŵp Astudiaeth Iraq ei adroddiad heddiw am 12:00 o'r nos.
Gemma 4 31B-it COMET 91.6	Cyflwynodd Grŵp Astudyraeth Iارق ei adroddiad am 12.00 GMT heddiw.
TranslateGemma 4B COMET 84.9	Mae Grŵp Astudiaeth Iraq wedi cyhoeddi report yno am 12.00 GMT heddin.

6.2 Welsh → English

Example 2.1 — devtest line 811 (large disagreement between systems)

Source	Mae llewod, llewpartiaid hela a llewpartiaid yn swil weithiau a byddwch yn eu gweld yn well â sbienddrych.
Reference	Lions, cheetahs and leopards are sometimes shy and you will see them better with binoculars.
Techiaith NLLB 1.3B fine-tunes COMET 78.0	Lions, hedgehogs and leopards are sometimes ugly and you'll see them better with binoculars.
Qwen 3.6 27B COMET 76.2	Lions, hunting leopards, and shy leopards are sometimes seen, and you will spot them better with binoculars.
TranslateGemma 27B COMET 75.4	Lions, lionesses, and lion cubs can sometimes be stealthy, and you might see them better with binoculars.
Gemma 4 31B-it (Q4) COMET 74.2	Lions, lionesses, and cubs are sometimes shy and are best viewed with binoculars.
NLLB-200 3.3B COMET 71.3	Lions, hunting leopards and leopards are sometimes shy and you can see them better with a mirror.
Gemma 4 31B-it COMET 59.5	Lions, expanding prides, and dormant prides are sometimes seen, and you will see them better with binoculars.
TranslateGemma 12B COMET 47.8	Wolves, wolfhounds, and Irish wolfhounds can sometimes be quite docile and you may find them to be quite friendly.
TranslateGemma 4B COMET 42.8	The leaves, flowers, and fruits will be at their best and easiest to see when they are fully mature.

Example 2.2 — devtest line 921 (large disagreement between systems)

Source	Mae mynd mewn cwch yn weithgaredd hamdden cenedlaethol yn y Ffindir, gyda chwch i bob saith neu wyth o bobl.
Reference	Boating is a national pastime in Finland, with a boat to every seven or eight people.
Qwen 3.6 27B COMET 93.2	Boating is a national pastime in Finland, with one boat for every seven or eight people.
Techiaith NLLB 1.3B fine-tunes COMET 91.0	Boating is a national leisure activity in Finland, with a boat for every seven or eight people.
NLLB-200 3.3B COMET 91.0	Boating is a national leisure activity in Finland, with a boat for every seven or eight people.
TranslateGemma 27B COMET 84.3	Going out in a boat is a popular leisure activity in Finland, with boats available for every seven or eight people.
TranslateGemma 12B COMET 76.4	Sauna bathing is a national pastime in Finland, with a sauna for every seven or eight people.
Gemma 4 31B-it (Q4) COMET 72.7	Going for a swim is a national leisure activity in Finland, with a sauna for every seven or eight people.
TranslateGemma 4B COMET 66.0	It is a national sport in Finland, where teams of seven or eight people compete.
Gemma 4 31B-it COMET 57.4	Cuddling is a national leisure activity in the Findir, with one cuddle for every seven or eight people.

Example 2.3 — devtest line 231 (all systems score well)

Source	Cychwynnodd y rhaglen am 8:30 p.m. amser lleol (15.00 UTC).
Reference	The program started at 8:30 p.m. local time (15.00 UTC).
NLLB-200 3.3B COMET 97.7	The program started at 8:30 p.m. local time (15.00 UTC).
Techiaith NLLB 1.3B fine-tunes COMET 96.6	The programme began at 8:30 p.m. local time (15.00 UTC).
Qwen 3.6 27B COMET 96.3	The program started at 8:30 p.m. local time (15:00 UTC).
Gemma 4 31B-it COMET 96.2	The program began at 8:30 p.m. local time (15:00 UTC).
Gemma 4 31B-it (Q4) COMET 96.2	The program began at 8:30 p.m. local time (15:00 UTC).
TranslateGemma 12B COMET 96.2	The program began at 8:30 p.m. local time (15:00 UTC).
TranslateGemma 4B COMET 96.2	The program began at 8:30 p.m. local time (15:00 UTC).
TranslateGemma 27B COMET 94.3	The programme started at 8:30 p.m. local time (3:00 p.m. UTC).

Common error patterns

Observations from manual spot checks of the outputs, not a systematic error annotation.

- Untranslated English words left in Welsh output, especially technical terms, names of organisations and rarer vocabulary.
- Wrong day or month names and incorrect or missing initial mutations (soft, nasal, aspirate) after prepositions, possessives and numerals.
- Literal word-for-word choices that are grammatical but unidiomatic, or that pick the wrong sense of an ambiguous word.
- Dropped words or clauses in long sentences — fluent output that silently omits part of the meaning, which surface metrics can miss.
- Occasional invented or garbled words (in the weaker systems) that look plausibly Welsh but are not real words.

7. Deployment and recommendations

Current deployment

Component	Details
Translation model	Qwen 3.6 27B (Q4_K_M) served by Ollama — replaced TranslateGemma 27B (Q8_0) on 23 October 2026 on the strength of this benchmark
API	FastAPI translation service on port 10083 (backend/server_ollama.py)
GPU memory	about 22 GB of GPU memory (TranslateGemma 27B used about 32 GB)
Public endpoint	https://dev.studio.bluemesh.ai/welsh
Web UI	https://salimkt.github.io/welsh-translate-poc/
Speech input	vLLM serving distil-whisper/distil-medium.en (English-only speech recognition)

In this benchmark the deployed Qwen 3.6 27B scored COMET 85.5 for English → Welsh and 89.9 for Welsh → English, at 2.01 and 1.19 seconds per sentence respectively.

The speech front-end uses an English-only Whisper model, so Welsh speech cannot currently be transcribed; spoken input works only for English → Welsh.

Recommendations

- **Techiaith NLLB 1.3B fine-tunes** has the highest mean COMET (87.9), 0.2 above the deployed Qwen 3.6 27B (1.8 on English → Welsh). If the gap exceeds the 1–2 point sampling noise and the licence and resource cost are acceptable, consider switching; otherwise keep Qwen 3.6 27B. Note: its licence (CC-BY-NC-4.0) does not allow commercial use.
- **Options to reduce GPU memory:** smaller TranslateGemma variants (12B, 4B) were not part of this run; score them before considering a switch.
- **Keep a human in the loop for official Welsh.** Even the best system makes mutation, terminology and omission errors; Welsh-language publications that fall under the Welsh Language Standards should be reviewed by a fluent Welsh speaker before release.

Suggested next steps

- Run a larger evaluation (500+ sentences) on in-domain text, e.g. Senedd proceedings and Welsh public-service corpora, to measure performance on the content the service will actually translate.
- Commission a human evaluation by Welsh speakers (adequacy and fluency ratings, error categories) on a subset, to calibrate what COMET differences mean for readers.

- Replace the English-only speech model with a multilingual Whisper model (or a Welsh fine-tune) so that Welsh speech can be transcribed and translated.
- Add glossary / terminology support (preferred translations for organisation names and official terms) and evaluate its effect on consistency.
- Re-run the benchmark on an otherwise idle GPU for reliable latency and throughput figures, including batched requests.

Appendix A. Full scores

System	Dir.	COMET	chrF++	BLEU	Median sent. COMET	s / sent.	Total s
Techiaith NLLB 1.3B fine-tunes	en→cy	87.33	66.44	46.34	88.08	2.414	241.4
Techiaith NLLB 1.3B fine-tunes	cy→en	88.38	75.45	58.69	90.19	2.234	223.4
NLLB-200 3.3B	en→cy	86.83	68.30	48.13	88.95	1.951	195.1
NLLB-200 3.3B	cy→en	88.82	76.76	60.80	90.10	0.499	49.9
Qwen 3.6 27B	en→cy	85.50	60.64	35.71	86.39	2.006	200.6
Qwen 3.6 27B	cy→en	89.85	74.62	55.00	90.34	1.185	118.5
TranslateGemma 27B	en→cy	84.27	53.49	25.34	85.95	2.033	203.3
TranslateGemma 27B	cy→en	88.98	71.06	49.58	88.76	1.149	114.9
Gemma 4 31B-it	en→cy	77.16	50.00	20.89	79.50	22.270	2227.0
Gemma 4 31B-it	cy→en	88.42	74.02	55.16	89.45	6.717	671.7
Gemma 4 31B-it (Q4)	en→cy	76.68	53.39	28.11	77.51	2.207	220.7
Gemma 4 31B-it (Q4)	cy→en	88.57	72.60	53.96	89.91	1.102	110.2
TranslateGemma 12B	en→cy	76.63	46.03	18.18	76.63	1.360	136.0
TranslateGemma 12B	cy→en	87.10	64.94	40.84	88.58	0.989	98.9
TranslateGemma 4B	en→cy	48.86	26.28	5.27	47.19	0.921	92.1
TranslateGemma 4B	cy→en	80.53	52.20	28.65	82.37	0.558	55.8

Table A1. All scores to two decimals. Total s is the wall-clock time for the whole sample in that direction.

Appendix B. Environment

Component	Version	Used for
Ollama	0.21.2	TranslateGemma, Gemma 4 (Q4) and Qwen runs (llama.cpp backend)
transformers	5.20	Gemma 4 int8 and NLLB runs
torch	2.14	CUDA inference
sacrebleu	2.6.0	BLEU and chrF++
unbabel-comet	2.2.7	COMET (Unbabel/wmt22-comet-da)
reportlab	5.1.0	this report
matplotlib	3.11.2	charts
GPUs	2× RTX PRO 6000 Blackwell	96 GB each; shared with production services

Table B1. Software and hardware. Scoring and report package versions are detected when the report is built.

Appendix C. Prompts

TranslateGemma prompt template (single user turn; {s}/{t} are the language names, {sc}/{tc} the codes, e.g. English (en) → Welsh (cy). The blank lines before the text are part of the template.

```
You are a professional {s} ({sc}) to {t} ({tc}) translator. Your goal is to
accurately convey the meaning and nuances of the original {s} text while
adhering to {t} grammar, vocabulary, and cultural sensitivities.
Produce only the {t} translation, without any additional explanations or
commentary. Please translate the following {s} text into {t}:


{text}
```

Generic system prompt used for the other LLMs (the sentence is sent as the user message):

```
You are a professional {s} to {t} translator. Translate the user's text into
natural, fluent, grammatically correct {t}, preserving meaning, tone, names
and numbers. Output only the translation, with no notes or explanations.
```

Line breaks inside the templates above are added for display; the prompts were sent as continuous sentences.